

Osaid Khan

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Dear Hiring Team,

It is with great interest that I submit my candidacy for the **Software Engineer - Full Stack** role with **Emergence AI**. I bring experience building backend services, frontend workflow tools, and production agent systems with **Python**, **FastAPI**, **React**, **TypeScript**, **PostgreSQL**, **Redis**, and **CI/CD**, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade backend and agentic systems that directly reflect **the technical depth this position demands**. As a Software Engineer at Sperse, I designed a multi-agent CRM orchestration system using **LangGraph**, **LangChain**, and **FastAPI** that routes requests to 40+ specialized sub-agents with secure access control and production **RESTful** and **event-driven APIs**. I also built **test automation** and regression suites across backend APIs, **frontend components**, and agent workflows by replaying OpenTelemetry traces in **GitHub Actions CI/CD** to validate **integration tests**, **end-to-end test scenarios**, and **automated testing frameworks** before **production deployment**. To support daily operations, I built a **React** Flow and **TypeScript** workflow UI for agent monitoring and workflow visualization, plus a **FastMCP** server exposing 160+ backend actions with **logging**, **tracing**, and **performance monitoring** and multi-tenant **observability dashboards**. I also built **LlamaIndex**-powered RAG agents with OpenAI embeddings and **Pgvector** to answer product questions and generate ready-to-run **REST** API examples for internal development. Prior to Sperse, as a Software Engineer at Qualitest, I designed a cloud-agnostic Storage Service in **Java** and **Springboot** with **PostgreSQL** and **Redis** that handled file operations across **Google Cloud Storage**, **S3**, and **MinIO** via expiring signed URLs. I also optimized **SQL** workloads in **PostgreSQL** with window functions, materialized views, and indexes, exposed the results through **React** visualizations for business stakeholders, and orchestrated long-running, event-driven jobs with **AWS Step Functions**, reducing **AWS Lambda** timeouts and improving error recovery across **100K+** weekly executions. Together, these roles have prepared me to contribute meaningfully to the team's core engineering responsibilities while owning features end-to-end.

Emergence's focus on **making mission-critical autonomy possible** resonates strongly with the way I like to build: verified, measurable, and reliable. Your emphasis on **verified autonomy** and **performance, safety, and reliability** aligns with the testing and instrumentation work I have already been doing across agent workflows, backend APIs, and production infrastructure. I would be excited to bring that mindset to **this team and its mission**, especially where your platform needs trustworthy execution, strong observability, and careful iteration under senior engineering guidance.

I would welcome the opportunity to discuss how my background can help advance **making mission-critical autonomy possible** for **Emergence AI**. I can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration.

Sincerely,
Osaid Khan